

Phase 3: what the lens search varies

Printed seven-core tip for the NV-diamond probe · generated 2026-07-21 20:05 from lens_design.py

1 Geometry searched for every lens family

These four are searched on a 1 μm integer grid. Everything else about the tip follows from them.

Parameter	Range (μm)	Step	What it sets
Air gap air_gap_um	5 to 250	1	closest approach of the printed tip to the diamond; also the central cap apex
Central post height central_height_um	5 to 200	1	IP-S pillar under the central (green) cap
Side post height side_height_um	5 to 120	1	IP-S pillar under each of the six collection caps
Side cap decentre side_core_offset_um	-30 to 110	1	cap centre relative to the 35 μm core pitch; negative moves it inward, which steers the collection cone toward the axis

2 Cap size, searched as a multiple of the beam

Cap radius is **not** searched directly. It is tied to the footprint the core's own full-NA beam throws onto it, so a cap can never be fitted over one region and used over a much larger one:

$$\text{footprint} = 2 \cdot w_{\text{mode}} + \text{height} \cdot \tan(\arcsin(\text{NA} / n_{\text{IP-S}}))$$
$$\text{cap radius} = \text{decentre} + \text{scale} \times \text{footprint}$$

Parameter	Range	Meaning
central_aperture_scale	1.0 to 1.35	1.0 exactly covers the beam; the upper limit is how far the cap may run past it
side_aperture_scale	1.0 to 1.35	1.0 exactly covers the beam; the upper limit is how far the cap may run past it

Consequence: the central/side overlap and the cap apertures are reported *results*, not free knobs. They follow from the post heights, the decentre and these two scales.

3 Surface shape, per lens family

Each surface starts as a Snell-law fit: for every point on the cap the code solves for the air direction that reaches the NV, takes the normal vector refraction demands, and least-squares fits the polynomial to those normals. The family then decides which coefficients the search may scale.

Family	Free shape parameters	Surface it can form
quadratic	radius_scale 0.35 to 2.5	one radius of curvature - a plain paraboloid cap
asphere	radius_scale 0.35 to 2.5 asphere_scale -1.0 to 3.0	one radius plus a rotationally symmetric fourth-order term
biconic	radius_x_scale 0.35 to 2.5 radius_y_scale 0.35 to 2.5 asphere_scale -1.0 to 3.0	independent radii along the radial and tangential axes, plus a fourth-order term
freeform	radius_x_scale 0.35 to 2.5 radius_y_scale 0.35 to 2.5 odd_scale -1.0 to 3.0 quartic_scale -1.0 to 3.0	independent radii, off-axis odd terms and an independent quartic - the least constrained surface

Every side cap additionally gets **tilt_scale** (0.35 to 2.5), which scales the vertex tilt that aims its cone at the axis. The central and side families are chosen independently, so all 16 pairings are searched.

4 Rules a candidate must satisfy

Checked before any ray is traced, so violating designs cost nothing and can never reach the results.

Rule	Value	Why
Print volume	$300 \times 300 \times 300 \mu\text{m}$	the two-photon printer's field
Cap covers its beam	scale ≥ 1.0	no cap fitted over one region and used over another
Cap not oversized	$\leq 1.35 \times$ footprint	beyond the fit the polynomial is unconstrained extrapolation
Cap owns its own light	$\geq 85\%$ of its lit footprint	otherwise its core is really refracting off a neighbouring cap
Apex is the real clearance	within $0.5 \times$ footprint	so the reported air gap is what the beam sees
Surface slope	≤ 3.0 ($\sim 72^\circ$)	printability; steeper walls also reflect light away
Polymer left under a cap	$\geq 5 \mu\text{m}$	a cap pinched to a sliver cannot be printed

5 How the search moves

Stage	Setting	Note
Family pairings	16	every central family against every side family
Restarts each	2	different seeds, best kept
Screening pass	differential evolution, 120 generations	the same objective at reduced fidelity - 3 depths, 1 wavelength, coarse grid
Full pass	differential evolution, 80 generations	9 depths, 9 wavelengths, 81^2 grid
Local refinement	coordinate descent, steps $10 \rightarrow 5 \rightarrow 2 \rightarrow 1 \mu\text{m}$	and $0.2 \rightarrow 0.02$ on the shape scales
Final selection	top 3 re-scored at full resolution	33 depths, 33 wavelengths, 161^2 grid - so the winner is chosen on the grid it is reported on

6 Held fixed (not searched)

Quantity	Value
Fibre NA / mode-field diameter	0.22 / $10 \mu\text{m}$
Core pitch	$35 \mu\text{m}$ (6 side cores + 1 central)
IP-S polymer index	1.52
Diamond index at the design wavelength	2.4058
Collection band	650 to 800 nm
Side-lens design wavelength	706.4 nm (emission-weighted centre of the band)
Excitation	532 nm, 10 mW out of the central core
NV layer	80 to $90 \mu\text{m}$ below the diamond surface
NV concentration	3 ppm

7 What is optimised

Each candidate is ray traced with exact vector Snell refraction and polarisation-averaged Fresnel at both the polymer-air and air-diamond surfaces, with total internal reflection removing the light that cannot escape the diamond. The signal is the green excitation rate multiplied by the six side cores' collection probability, integrated through the NV layer - only an NV that is both pumped and seen contributes. The search minimises shot-noise-limited magnetic sensitivity, which rewards more collected light and a tighter sensing spot at the same time.

Collection efficiency, quoted separately in the comparison figures, is a pure optical result of the same trace: it needs no assumed contrast, no linewidth model and no empirical normalisation.